

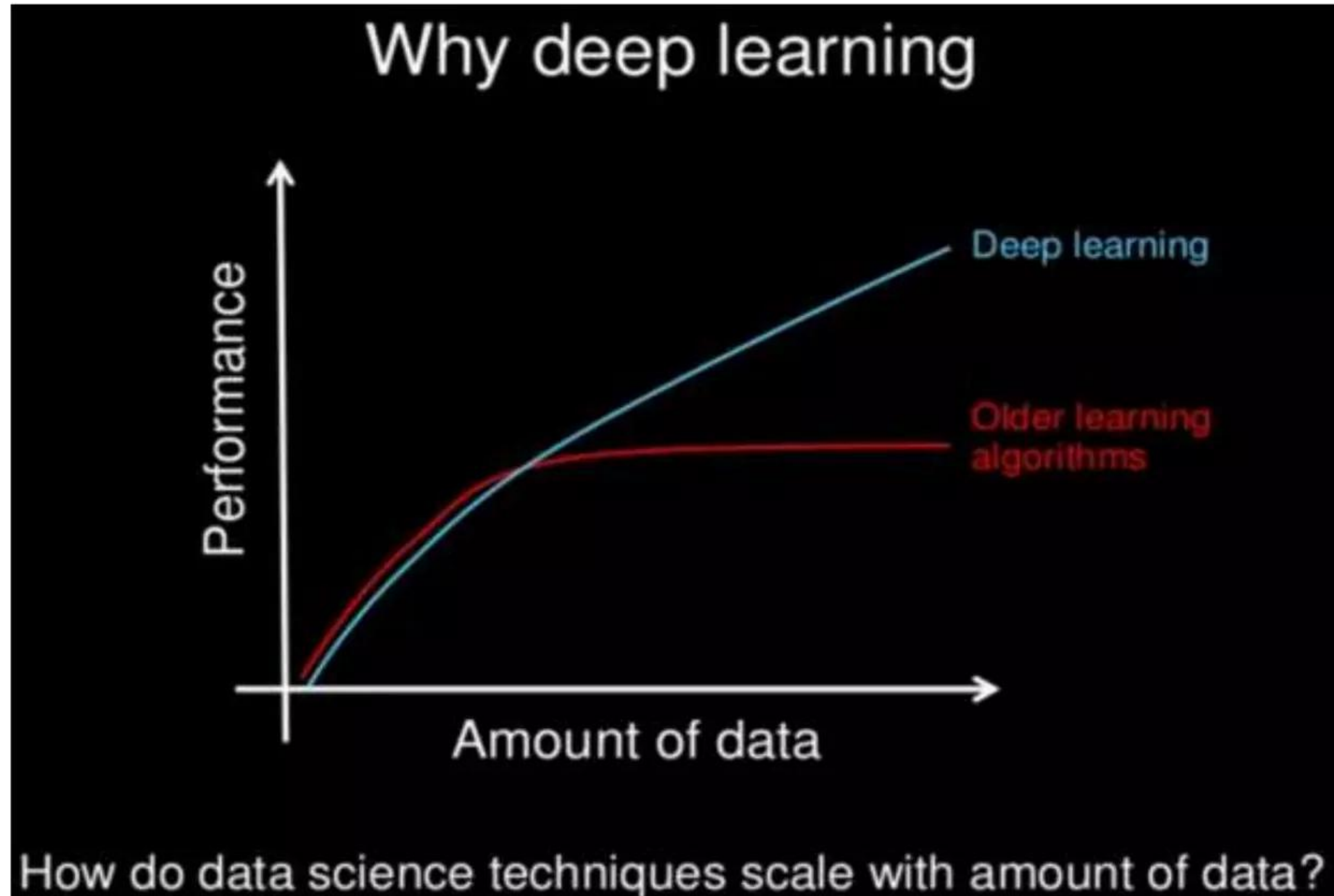
Deep Learning With Neural Networks

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Deep Learning

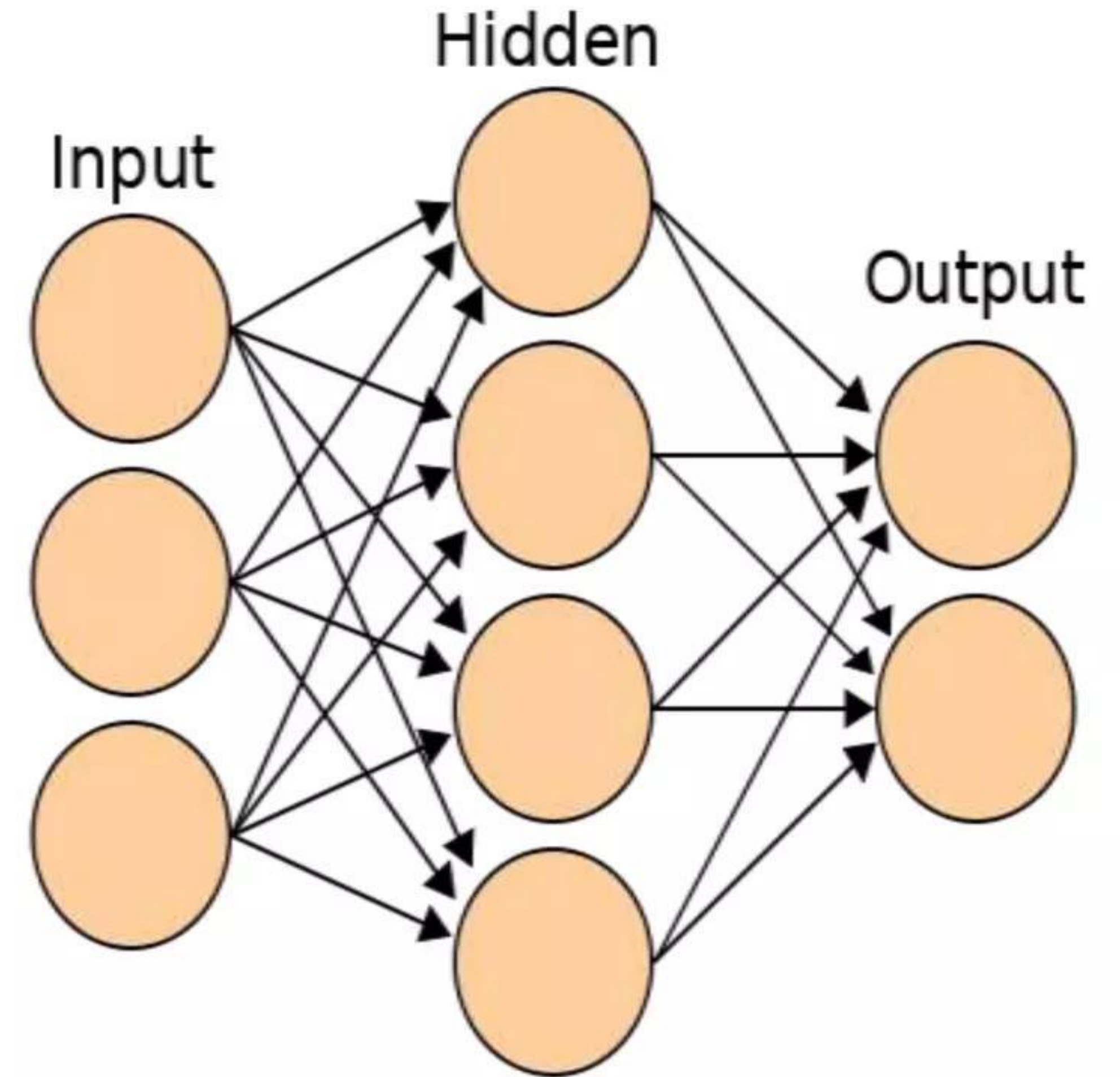
The term Deep Learning refers to training Neural Networks and sometime very large Neural Networks.

Deep Learning algorithms can learn very complex patterns, if provided enough data.



What is Neural Networks?

- Neural Networks are computing systems inspired by Biological Neurons. They try to mimic the functionality of the brain.
- It is constructed of neurons which learns different patterns and each neuron together produce the correct output.
- Each Neurons has weight which are initialized with random value and the values are updated to catch a pattern by a series of propagations through the network.



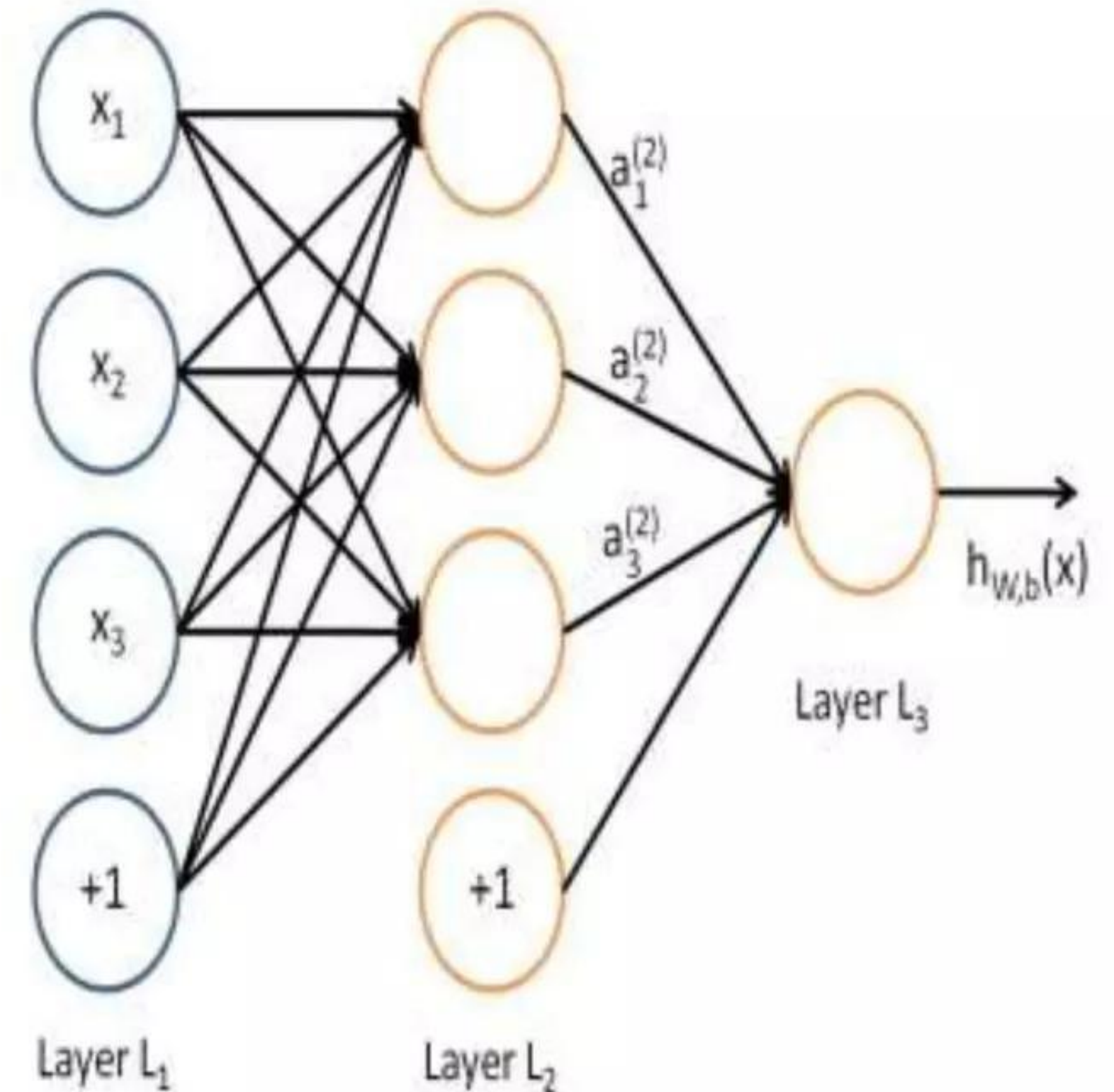
❑ Layers in a neural networks architecture:

- Input layer: The external information or data is fed in the units of input layer.
- Output layer: This layer gives the final output or prediction of the data fed into input layer.
- Hidden layer: Layers except the input and output layers are called Hidden layers. These are intermediate layers. There can be many hidden layers in a Network.

❑ Propagations in Neural Networks

Forward Propagation:

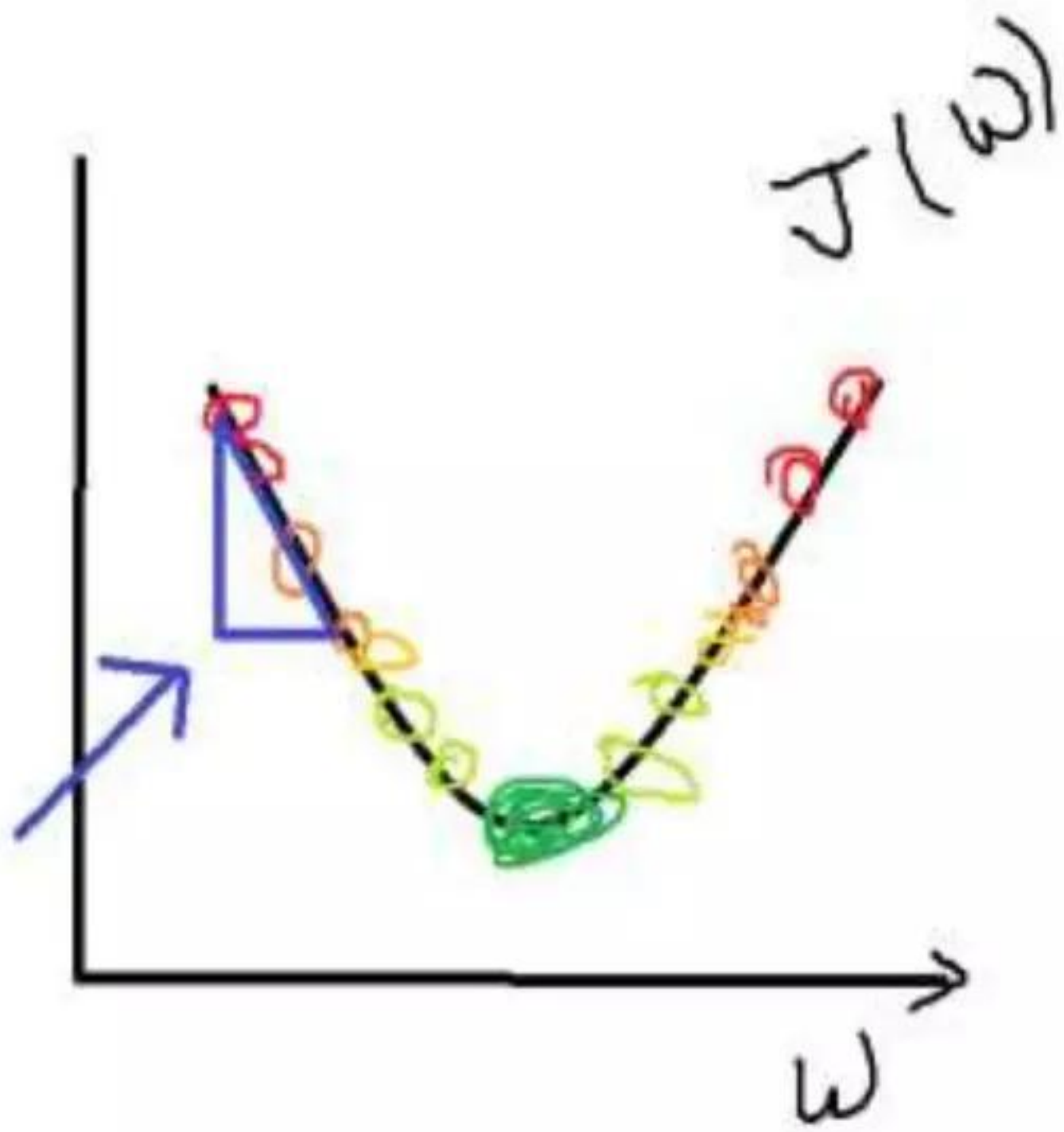
- As the name suggest, a series of calculations are performed in neuralNets from first to the last layer.
- The weights are multiplied with input and passed through an activation function, the value obtained is passed to the next layer and the same step is repeated until the output layer.
- The value of output layer is used to calculate the error in the model.



❑ Back-Propagation

- It is the main reason a neural network work.
- The randomly initialized weights are updated to minimize the error in the prediction. The updation starts from last layer to the first hidden layer.
- The weight updation is done using the *Gradient Descent* algorithm.

❑ Gradient Descent Algorithm



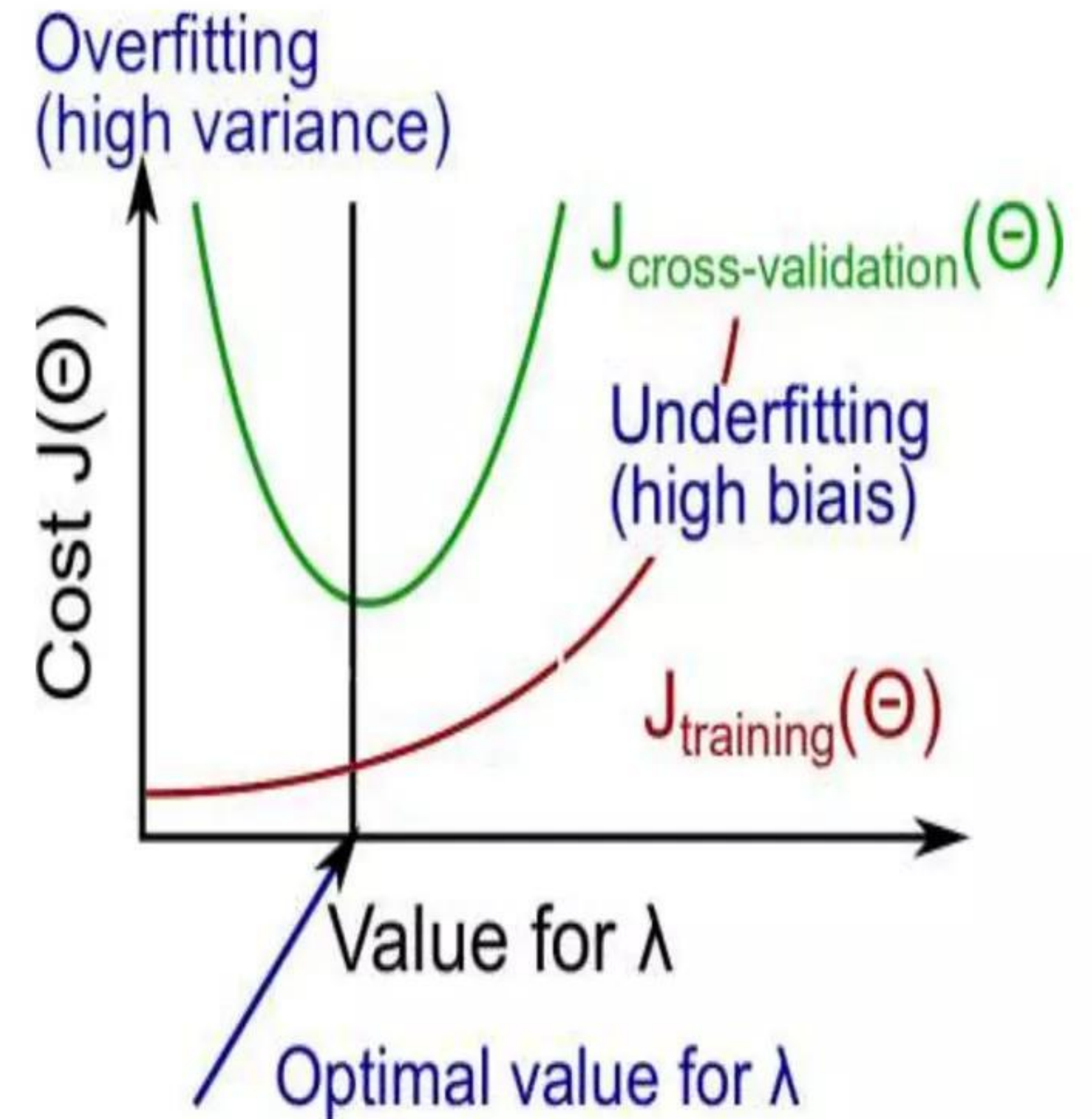
$$\begin{array}{l} \text{Repeat} \\ \left\{ \begin{array}{l} w := w - \alpha \frac{dJ(w)}{dw} \end{array} \right. \\ \end{array}$$

The **gradient descent** algorithm calculates the error derivative w.r.t. weight and it is multiplied by the learning rate and then subtracted from the old weight. The gradient descent algorithm has been optimized and modified in recent years by many researchers. Some of the optimized learning algorithms are RMSProp and Adam.

❑ Bias-Variance Problem

During the training of Neural Network algorithms the model can suffer overfitting and underfitting.

- **Overfitting:** When the model performs well on training set but fails to generalise on the validation set, is called *Overfitting*. This usually happens when the architecture is large and the data is not sufficient. It is also called **High Variance**.
- **Underfitting:** It is the poor performance of model on the training set itself. It is also called **High Bias**.



❏ Controlling Overfitting and underfitting

Two main techniques to avoid overfitting problems are:

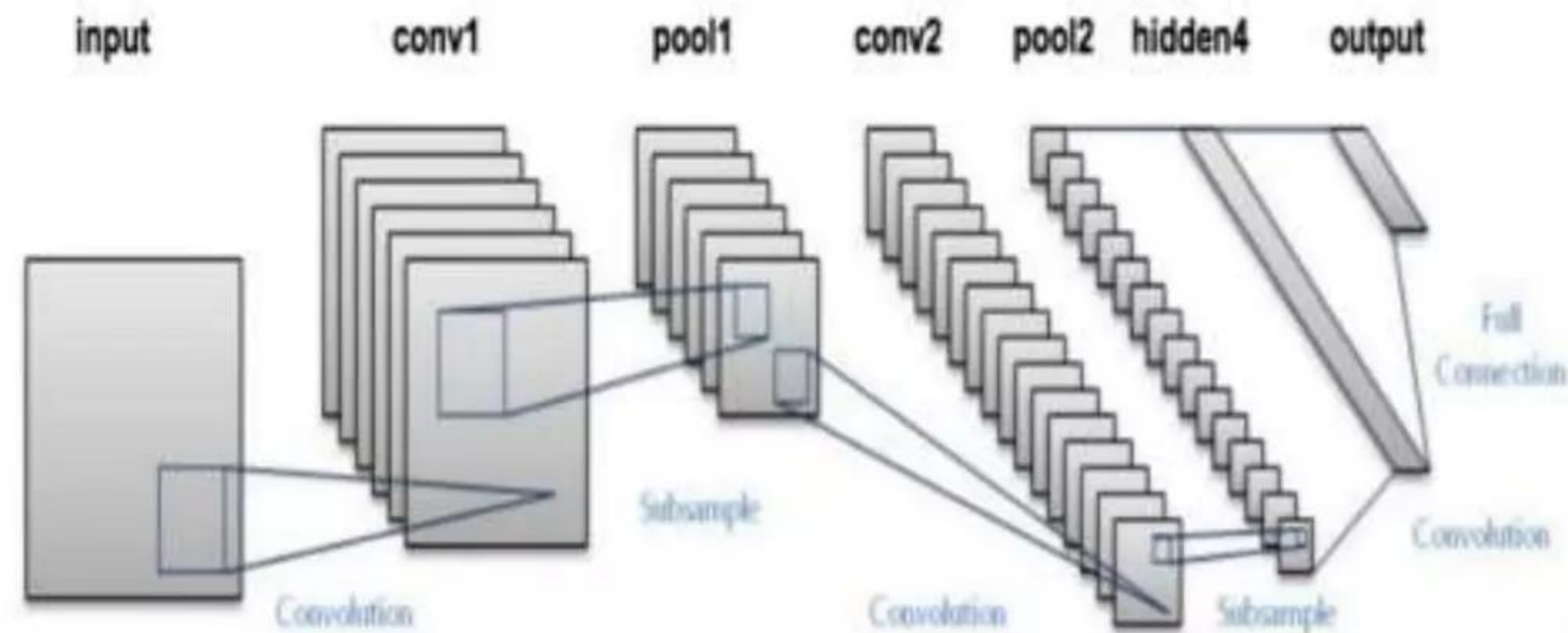
- **Regularization:** It is a technique used to reduce overfitting. Its basic concept is to penalize the weights. It comes in action with Gradient Descent algo.
- **Dropout:** In this technique the random neurons are disabled. It improves the model accuracy significantly.

To control underfitting:

- Number of hidden layers can be increased. Number of units of neurons in a layer can also be increased.

Convolutional Neural Networks (CNN)

- CNNs are widely used to solve computer-vision related problems.
- It can be used for variety of tasks like image classification, face recognition, style transfer etc.
- CNNs are formed of mainly 3 components: Convolution layer, Pooling layer and Fully connected layer.
- In 1989, Yann LeCun used back-propagation to train a CNN to classify digits. It became a foundation for modern computer-vision.



❑ Applications of CNN

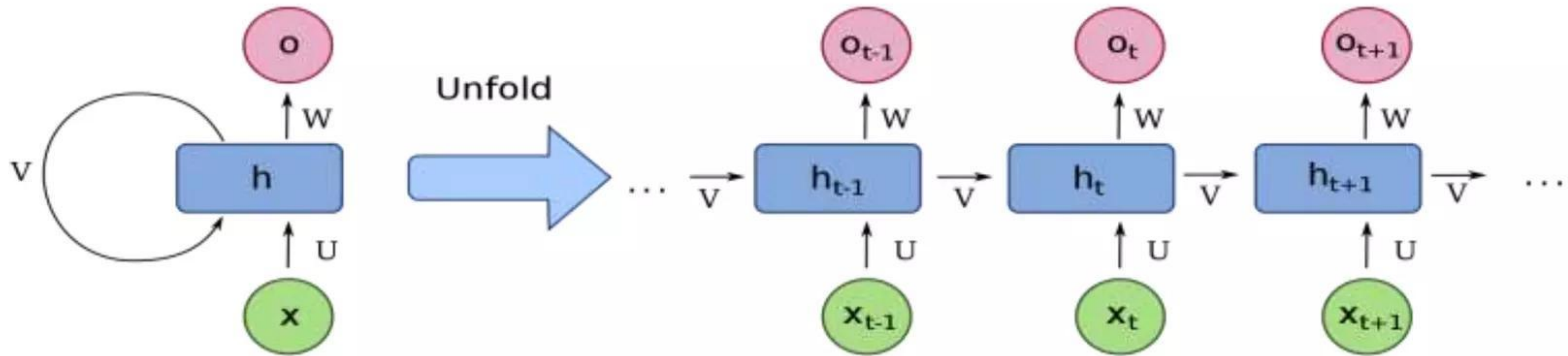
Image Processing: CNN has changed the traditional image processing methods. Using CNN face recognition and pattern matching can be done more accurately and with less computation power. It can be used for lossless compression of images.

Autonomous Systems: CNN has powered autonomous systems. Autonomous car, security cameras etc are not just part of sci-fi movies.

Style Transfer: CNNs are being used to create art.

Medical Image Computing: CNNs are being used in medical image computing to diagnose diseases. Recently, a model has been developed that performed better than Radiologists to diagnose Pneumonia from chest X-ray images.

Recurrent Neural Networks (RNN)



- RNNs are used to train sequential data like text and audio.
- RNNs is mostly used in Natural Language processing for recommendation systems and sentiment analysis.
- The main advantage of RNN is that it can remember contexts.

❏ Conclusion

- With availability of huge amount of data Neural Networks can be used to train complex patterns.
- Many groundbreaking algorithms have been implemented in recent years which can be used to solve many problems which the traditional methods could not.
- There are many applications of Neural Network algorithms which can help solve real life problems like in health sector and electric power consumption.
- Sequential models like RNN, LSTM and GRU are used to create chatbots.

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